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The Editor
Proceedings of the Royal Society A:
Mathematical, Physical and Engineering Sciences
The Royal Society
6–9 Carlton House Terrace
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Dear Editor,

We wish to submit the enclosed manuscript, *Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells*, for consideration as a Research Article in *Proceedings of the Royal Society A*.

The paper advances a unifying thesis for the mechanics of fluid-filled viscoelastic shells: shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem. These three roles — excitation filtering, spectral organisation, and inverse identifiability — have previously been studied in isolation across disparate application domains. The present work establishes them as logically connected consequences of the same geometric structure.

The manuscript contributes five formal results within a reduced analytical framework:

1. **Theorem 1 (Rank Collapse).** Equivalent-radius reductions of non-spherical fluid-filled shells are generically rank deficient: collapsing independent geometric parameters into a single scalar size descriptor confounds geometry and material stiffness, rendering independent parameter recovery impossible.
2. **Theorem 2 (Identifiability Lifting).** Oblate asphericity lifts this rank deficiency by introducing curvature-dependent mode splitting that provides the additional spectral information needed for locally identifiable inversion.
3. **Proposition (Near-Spherical Asymptotics).** The transition to the spherical limit follows a regular expansion, $\sigma_3(\varepsilon) = \lambda_1\varepsilon^2 + O(\varepsilon^4)$, confirming that the exact sphere is a singular point in the inverse problem and that any previously reported finite identifiability floor is a discretisation artefact.
4. **Lemma (Forward $\neq$ Inverse Adequacy).** A counterexample demonstrates that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere reduction

may predict resonant frequencies within engineering tolerance yet fail catastrophically when used for parameter estimation.

5. **Proposition (Breit–Wigner Absorption).** The energy fraction absorbed by the shell from an incident acoustic field follows a Breit–Wigner resonance profile, providing a rigorous basis for the excitation-filtering role of geometry.

The formal results are supported by cross-application evidence drawn from three distinct physical systems: the human abdomen (infrasound resonance and whole-body vibration coupling), the watermelon (acoustic ripeness assessment via shell stiffness gradients), and the urinary bladder (fill-dependent resonance tracking). Additional supporting evidence spans gas-pocket transduction, scaling-law analysis, borborygmi modelling, and sub-bass perception pathways.

This capstone paper synthesises the results of nine companion papers developed within the Browntone research programme, each of which addresses a specific aspect of the geometry–spectrum–inference nexus. We believe the unifying framework, the formal inverse-problem results, and the breadth of cross-application evidence make the manuscript well suited to the interdisciplinary readership of *Proceedings of the Royal Society A*.

The manuscript has not been published previously and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission to *Proceedings of the Royal Society A*.

We confirm that all necessary ethical approvals have been obtained. The study is purely analytical and computational and involves no human or animal subjects.

We have included suggested reviewers with expertise spanning inverse problems in structural dynamics, shell vibration, spectral geometry, and biomedical acoustics.

Yours sincerely,

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